

ASTEROIDS

Asteroid Subsurface Terrain Exploration Radar for Orbit Ice Detection
Survey

Spring 2026 – 16.831 DT-2(b): Revised Proposal for Flight Demo



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1 Executive Summary

1.1 Technical Overview

The **ASTEROIDS** project is a 2U CubeSat-class flight demonstration designed to emulate key aspects of asteroid resource prospecting – specifically, the detection of subsurface water-ice deposits on small bodies using radar and ultrasonic sensors. These volatiles are compelling because they represent potential propellant and life-support feedstocks for future deep-space missions. The 2023–2032 Planetary Science Decadal Survey ranks ISRU (In-Situ Resource Utilization) resource characterization as a top priority, and no mission has yet mapped Trojan asteroid subsurface ice at high resolution [1]. ASTEROIDS is designed as the first step toward filling that gap.

Our team will design, integrate, and operate a **2U CubeSat analog** ($10 \times 10 \times 22.7$ cm, <2 kg) equipped with spring-deployable radar antenna panels and an ultrasonic depth-sensing payload. Mounted on the AeroAstro zipline testbed, the analog CubeSat will simulate an orbital pass over a Trojan asteroid surface. The craft will autonomously deploy its sensing panels mid-flight via a **spring-driven mechanism** with a **servo motor backup**, confirm deployment via an onboard Hall-effect sensor, collect subsurface reflection data from buried analog targets representing water-ice layers, and transmit data to a laptop ground station in real time over a 915 MHz LoRa link.

This demonstration meaningfully addresses stakeholder goals by validating three key sub-systems:

- A **fold-out radar antenna mechanism** (spring primary, servo latch backup, Hall-effect confirmed), with deployment estimated at <0.18 s, well within the 3 s requirement;
- A **dual-sensor detection system** combining a short-range radar module (HB100, 10.525 GHz) and HC-SR04P ultrasonic transducer (40 kHz, 3.3V–5V compatible) for complementary subsurface echo profiling; the standard HC-SR04 is held as a backup unit;
- A **miniaturized telemetry and data-logging pipeline** (Adafruit RFM95W LoRa at 915 MHz) providing a link margin of >95 dB at 10 m range, with redundant SD card fallback.

Quantitative success criteria are defined in Section 4: 100% deployment success over ≥ 3 runs, $>80\%$ packet delivery, and depth measurement error $<10\%$. Subsystem analyses, power budgets, and data budgets are detailed in Section 5.

1.2 Project Plan Overview

Design and integration will take place from February to April 2026. The team’s efforts align individual learning objectives with functional subsystems while deliberately creating **cross-subsystem peer learning opportunities** at each integration milestone:

- **Zaire Williams** – Electronics and Integration: leads power distribution, sensor wiring, and hardware-software interfacing. Aims to independently design and verify a functional mixed-signal avionics system. Mentors Jorge on hardware timing constraints that inform embedded code design.
- **Jorge Cavazos** – Flight Software and Telemetry: develops real-time data-acquisition firmware on the Raspberry Pi Pico, focusing on interrupt-driven timing, packetization, and radio transmission reliability. Pairs with Andras during all end-to-end link tests.
- **Jonah Goldstein** – Mechanical and CAD Systems: designs the 2U CubeSat structure and deployable antenna assembly in SolidWorks, learning tolerance analysis and dynamic

hinge kinematics. Co-reviews harness routing with Zaire to prevent mechanical-electrical interference.

- **Andras Beleznyay** – Communications and Mission Operations: configures primary and backup LoRa links, validates the RF link budget, and manages ground-station Python scripts. Trains all members on the ground-station interface before flight day.

Weekly cross-subsystem reviews (30 min, rotating lead) ensure every team member develops breadth beyond their primary role. Following fabrication and validation, the final demonstration will occur using the 22 m zipline between April 13–29. Total resources amount to approximately \$200 and 120–150 post-spring-break person-hours.

2 Concept of Operations (CONOPS)

The ASTERIODS zipline demonstration simulates a CubeSat performing a low-altitude orbital pass over a Jupiter L5 Trojan asteroid. The analogy mapping is direct: spring-deployable panels emulate low-gravity instrument deployment; buried foil/foam targets emulate subsurface asteroid ice layers; the 915 MHz LoRa link emulates a spacecraft downlink; and SD card fallback emulates DSN contingency data-return procedures [7]. This operational philosophy mirrors the approach taken by small-body missions such as OSIRIS-REx and Hayabusa2 [2, 3].

Figure 1 presents the **dynamic CONOPS flowchart**, showing all mission phases with their trigger conditions and contingency branches. Figure 2 shows the physical zipline geometry.

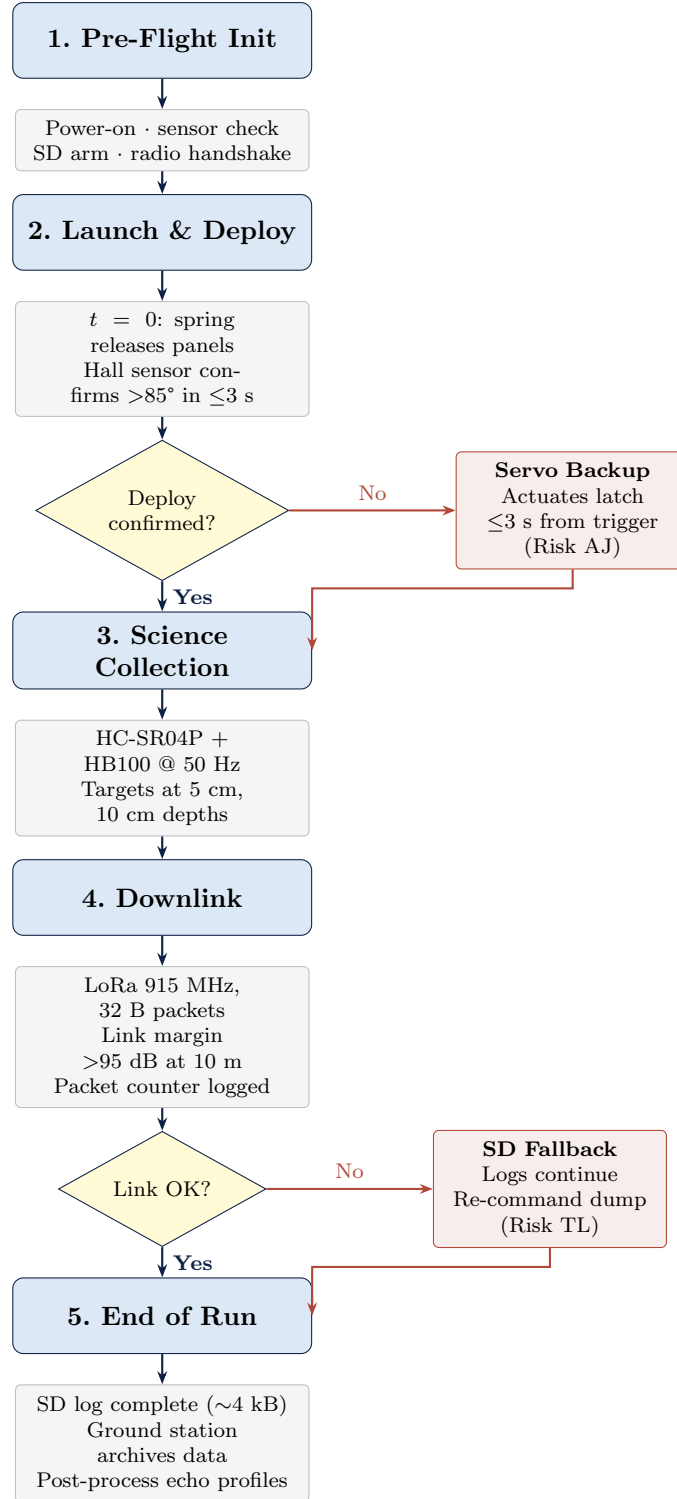


Figure 1: Dynamic CONOPS flowchart for the ASTEROIDS zipline demonstration. Blue boxes show primary mission phases with key parameters; red boxes show contingency branches. Every phase has a measurable trigger condition and maps to at least one risk-register item.

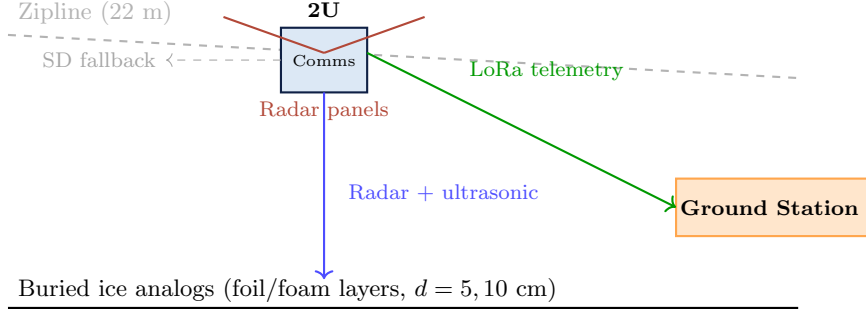


Figure 2: Physical layout of the ASTERIODS zipline test geometry. Buried analog targets at known depths enable direct ground-truth comparison with sensor echo profiles.

1. Initialization / Pre-Flight Verification. Before launch, the 2U CubeSat analog is powered on and undergoes rapid diagnostics (~ 30 s). A serial handshake confirms the LoRa radio link, SD-card write speed, and sensor readiness. The system enters low-power standby (≈ 0.08 W) until the launch command.

2. Launch and Panel Deployment (Spring Primary, Servo Backup, Hall-Effect Confirmed). At $t = 0$ s, the zipline carriage releases. A pre-loaded torsion spring ($k \approx 5$ N/m, pre-compression $x = 5$ cm) stores $E_{\text{spring}} = \frac{1}{2}kx^2 \approx 6.3$ mJ. Rotational energy analysis (Section 5) estimates each 50 g panel reaches $>90^\circ$ in ≈ 0.18 s, well within the 3 s requirement (R1.1). A Hall-effect sensor confirms the achieved angle. If confirmation is not received within 3 s, a servo motor (5 V) actuates the latch as a backup (Risk AJ).

3. Science Collection Phase. Once panels are confirmed deployed, the Pi Pico triggers both the HC-SR04P (40 kHz ultrasonic, 3.3V–5V compatible) and HB100 (10.525 GHz radar) at a combined 50 Hz sample rate. Subsurface targets (layered copper foil and foam sheets at 5 cm and 10 cm depth) produce time-of-flight echoes that distinguish material layers. All data is time-stamped and logged to the SD card (~ 4 kB per 10 s run).

4. Downlink Operations. Telemetry packets (timestamp, radar echo amplitude, ultrasonic ToF, RSSI, sequence number) are downlinked via the primary Adafruit RFM95W LoRa transceiver at 915 MHz. The link budget (Section 5) yields a link margin of >95 dB at 10 m – far exceeding the 10 dB requirement (R3.1). A second RFM95W on a separate channel remains in warm standby.

5. Data Processing and Interpretation. Received data is filtered, synchronized by sequence number, and converted into reflection-amplitude vs. time profiles. Correlation with known target positions verifies depth accuracy (error $<10\%$), and post-processing identifies distinct material echo peaks (R7).

6. Contingency and Recovery. If telemetry is lost, SD logging continues uninterrupted (Risk TL). A post-run commanded data dump recovers complete records, directly analogous to DSN contingency procedures used on deep-space CubeSats [7].

3 Risk Analysis and Mapping to Test Campaign

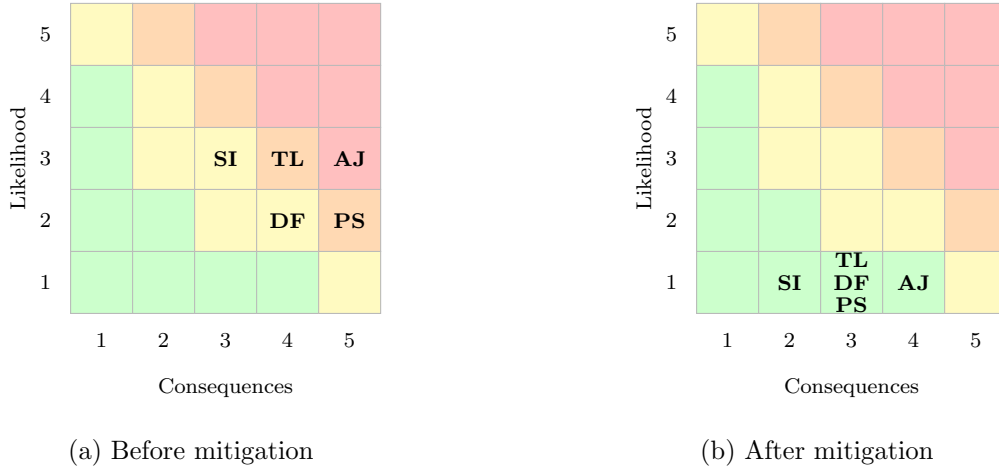


Figure 3: Risk matrices (Green - safe to occur, Yellow - beware of complications, Red - mission critical)

The rogue operand classification ensures each risk is traced to a physical causal mechanism rather than described phenomenologically, following Leveson's STAMP framework [5]. Every high-priority risk maps to an explicit fault-injection test in the demonstration.

Table 1: Risk Register with Rogue Operand, Mitigation, and Demo Verification

ID	Risk	Rogue Op.	L	C	Mitigation	L'	C'	Demo Verification
AJ	Antenna jams on deploy	Matter	3	5	Pre-flight hinge cycling (≥ 10 cycles); spring torque $3\times$ required; servo actuates if Hall sensor does not confirm $>85^\circ$ within 3 s.	1	4	Intentionally halt spring latch; verify servo opens latch and Hall sensor confirms within 3 s.
TL	Data transmission loss	Info.	3	4	Dual RFM95W LoRa radios on separate channels; SD card logs all data at 50 Hz regardless of link status.	1	3	Disable primary radio mid-run; verify SD log is complete; verify backup link or commanded dump recovers all packets.
DF	Target detection failure	Info.	2	4	Calibrate HC-SR04P at 5 and 10 cm using ruler ground truth; require SNR >6 dB; standard HC-SR04 held as backup unit.	1	3	Verify distinct echo peaks for two target materials in controlled layout; depth error $<10\%$.
SI	Structural instability	Forces	3	3	SolidWorks FEA at $2\times$ estimated zipline acceleration; CoM within 2 cm of center; shock-resistant hinge joints.	1	2	Full-speed zipline run; inspect for detachment, oscillation, or loss of contact.
PS	Power failure	system Elec. Energy	2	5	1200 mAh Li-ion verified $>90\%$ capacity pre-run; $5\times$ runtime margin.	1	3	Disable non-critical rail mid-run; confirm mission continues; thermal check at peak load.

4 Testing and Demonstration Description

4.1 System and Subsystem Requirements

Table 2: High-Level (L1) System Requirements

ID	Category	Requirement	V&V
R1	Functional	The CubeSat analog shall autonomously deploy radar panels mid-flight via spring (primary) with servo backup; deployment shall be Hall-effect confirmed within 3 s.	Analysis + Test
R2	Performance	The payload shall detect reflections from buried targets >2 cm thick at 5 cm and 10 cm depths, with depth error <10%.	Test
R3	Performance	Data shall be stored locally and transmitted with >80% packet success at ≤ 10 m range.	Analysis + Test
R4	Environmental	The system shall survive full-speed zipline motion without damage or detachment.	Test + Sim
R5	Constraint	Total mass <2 kg.	Analysis
R6	Performance	Battery shall support >1.5 h continuous operation at full sensor and radio load.	Analysis + Test
R7	Performance	Post-processing shall identify distinct echo profiles for ≥ 2 different buried target materials.	Test + Sim
R8	Constraint	Total volume within 2U envelope ($10 \times 10 \times 22.7$ cm).	Analysis

4.2 Verification Approach

Table 3: Subsystem (L2) Derived Requirements and Verification

ID	Parent	Category	Derived Requirement	V&V
R1.1	R1	Functional	Spring deployment shall rotate panels $>90^\circ$ within 3 s; Hall-effect angle sensor confirms.	Test + Analysis
R1.2	R1	Functional	Servo backup shall actuate latch within 3 s of no Hall confirmation.	Test
R1.3	R1	Functional	Hinge joints shall survive ≥ 10 deployment cycles without degradation.	Test
R2.1	R2	Performance	HC-SR04P (primary) shall measure 5 cm and 10 cm targets within ± 1 cm error; HC-SR04 (backup) shall meet the same criterion if substituted.	Analysis + Calib.
R2.2	R2	Functional	Software shall apply 5-point moving-average baseline subtraction; SNR > 6 dB for valid detection.	Test
R3.1	R3	Performance	LoRa link shall achieve $> 80\%$ packet delivery at ≤ 10 m (line-of-sight).	Analysis + Test
R3.2	R3	Performance	End-to-end latency from sensor acquisition to logged packet shall be < 5 s.	Analysis + Test
R4.1	R4	Environmental	CubeSat shall maintain zipline contact at full speed; no part detachment.	Test
R4.2	R4	Environmental	Structure shall pass static load test at $2\times$ estimated peak zipline acceleration.	Analysis + Test
R5.1	R5	Constraint	All components shall not exceed 2 kg total mass (target: < 1.5 kg with 25% margin).	Analysis
R6.1	R6	Performance	Battery shall retain $> 90\%$ rated capacity; estimated runtime > 7 h at average 1.1 W load.	Test
R6.2	R6	Performance	Battery temperature shall remain within $0-45^\circ\text{C}$ during full load.	Analysis + Test
R7.1	R7	Performance	Post-processing shall show distinct echo peaks for foil vs. foam targets (> 3 dB amplitude difference).	Analysis + Test
R7.2	R7	Performance	Depth profile shall correctly identify both target depths in ≥ 3 of 3 demonstration runs.	Analysis + Test
R8.1	R8	Constraint	All components shall fit within $10 \times 10 \times 22.7$ cm with panels stowed.	Analysis

4.3 Test Campaign and Expected Results

Testing proceeds in three phases, designed to de-risk each fault scenario in the Risk Register with quantitative pass/fail criteria.

Phase I — Subsystem Validation:

- *Antenna Deployment Test (Risk AJ; R1.1–R1.3):* 10-cycle repeatability; record time-to-deploy and Hall-confirmed angle. Fault injection: hold spring latch and verify servo backup actuates within 3 s.
- *Sensor Calibration (R2.1–R2.2):* Static HC-SR04P measurements at 5, 10, 15, 20 cm against ruler; generate calibration curve confirming < 1 cm error. Apply baseline subtraction and verify SNR > 6 dB. Repeat with the backup HC-SR04 (requires a 5V supply and voltage divider on the echo pin to protect the Pi Pico’s 3.3V GPIO) to confirm it meets the same accuracy requirement.

- *Power System Test (Risk PS; R6.1–R6.2)*: 1-hour continuous load simulation; measure voltage drop <5%; verify thermal compliance.
- *Radio and Logging Test (Risk TL; R3.1–R3.2)*: Validate >80% packet delivery at 10 m; power off primary RFM95W mid-stream and confirm SD log continuity; verify backup channel activates.

Phase II — Environmental and Integration Tests:

- *Vibration/Shock Test (Risk SI; R4.1–R4.2)*: Manual impulse test; verify no hinge binding or structural loosening. SolidWorks FEA at $2\times$ estimated acceleration.
- *Thermal Test*: 0°C cold chamber test for battery and sensor stability over 30 min (R6.2).
- *Integrated Operations Test*: All subsystems active simultaneously; verify no power faults or system resets; confirm baseline subtraction behaves correctly (R2.2).

Phase III — End-to-End Zipline Demonstration:

Full 22 m flight profile with live telemetry and SD logging. Targets at 5 cm and 10 cm are placed at positions not disclosed to the ground-station operator; the processed echo profiles must correctly identify both depths.

Integrated demo verification (risk-register aligned):

- *AJ*: Block spring latch; verify servo opens within 3 s with Hall sensor confirmation.
- *TL*: Disable primary RFM95W; verify complete SD log and backup link recovery.
- *DF*: Controlled target layout; verify distinct peaks for foil vs. foam.
- *SI*: Full-speed run; inspect for detachment and oscillation.
- *PS*: Disable non-critical rail; confirm mission continues; check temperatures.

Quantitative success criteria:

- Deployment success rate = 100% across ≥ 3 flights (including 1 fault-injection run).
- Packet delivery >80%; end-to-end latency <5 s.
- Depth error <10% for both 5 cm and 10 cm targets.
- No structural failure or detachment across all runs.

5 Subsystem Engineering Analyses

5.1 Communications Subsystem — RF Link Budget

The Adafruit RFM95W uses the Semtech SX1276 LoRa transceiver at 915 MHz. The link budget is computed from the Friis transmission equation [6]:

$$P_r = P_t + G_t + G_r - \text{FSPL}$$

Free-space path loss at $f = 915$ MHz, $d = 10$ m:

$$\text{FSPL} = 20 \log_{10}(d) + 20 \log_{10}(f) + 20 \log_{10}\left(\frac{4\pi}{c}\right) = 20 + 179.2 - 147.6 \approx 51.7 \text{ dB}$$

With $P_t = +20$ dBm, $G_t = G_r = +2$ dBi (spring monopole antennas, 3 total: 2 for CubeSat, 1 for ground station), and RFM95W receiver sensitivity -123 dBm at SF7/BW125 kHz:

$$\text{Link Margin} = 20 + 2 + 2 - 51.7 - (-123) = \mathbf{95.3 \text{ dB}} \gg 10 \text{ dB required (R3.1)}$$

This margin is deliberately conservative and provides robustness against body-shadowing and polarization mismatch in the outdoor zipline environment. At maximum sensitivity (SF12), the RFM95W reaches -148 dBm, pushing the theoretical margin even higher.

Redundancy: Two independent RFM95W transceivers on separate channels (procured at \$29.95 each). An Adafruit Micro SD SPI breakout board logs all data locally at 50 Hz at all times, providing complete mission data under total radio failure.

5.2 Payload Subsystem — Sensor Analysis and Data Budget

Ultrasonic depth sensing (HC-SR04P, primary): The HC-SR04P operates at 40 kHz with a 3.3V–5V supply range and 3.3V-compatible logic levels, making it directly compatible with the Raspberry Pi Pico GPIO without a voltage divider. Range: 2–400 cm; resolution: ~ 3 mm. Time-of-flight to target at depth d :

$$\Delta t = \frac{2d}{v_{\text{sound}}} \approx \frac{2 \times 0.10}{343} \approx 0.58 \text{ ms} \quad (10 \text{ cm target})$$

This is well within the sensor’s 38 ms echo window. The standard **HC-SR04** is procured as a backup unit; if substituted, a resistor voltage divider ($1 \text{ k}\Omega + 2 \text{ k}\Omega$) must be added on the echo line to step the 5V output down to 3.3V before the Pi Pico GPIO pin.

Radar module (HB100): CW Doppler module at 10.525 GHz, ≈ 100 mW. In CW reflection mode, Doppler shift from stationary targets is zero but amplitude variation detects subsurface dielectric contrast. The relative permittivity step between foam ($\epsilon_r \approx 1.05$) and copper foil ($\epsilon_r \gg 1$) produces a measurable reflected power change that the baseline subtraction algorithm resolves as a distinct echo peak.

Data budget: 50 Hz sampling for a 10 s run yields 500 samples. Each sample = 8 B (4 B radar amplitude + 4 B ultrasonic ToF). Total raw data: ≈ 4 kB per run, far within SD capacity and LoRa bandwidth at SF7.

5.3 Power Subsystem — Mode Budget

The Raspberry Pi Pico microcontroller operates at 3.3 V and is powered via a 5 V buck converter (LM2596, $\eta \approx 92\%$). A dedicated 3.3 V LDO (AMS1117) feeds the RFM95W modules. Table 4 summarizes the peak science collection mode power draw.

Table 4: Power Budget — Science Collection Mode (Peak)

Subsystem	Component	Current (mA)	Power @ 3.3/5V (W)
Logic	Raspberry Pi Pico	25	0.083
Payload	HC-SR04P (ultrasonic)	15	0.075
Payload	HB100 radar module	100	0.33
Comms	RFM95W primary (Tx)	100	0.33
Comms	RFM95W backup (standby)	5	0.017
Storage	Micro SD card (active)	100	0.33
Mechanisms	Servo (standby)	5	0.017
Total		350	1.18 W

Battery sizing: 1200 mAh Li-ion at 7.4 V = 8.88 Wh. At 1.18 W average:

$$t_{\text{run}} = \frac{8.88 \text{ Wh}}{1.18 \text{ W}} \approx 7.5 \text{ h} \gg 1.5 \text{ h requirement (R6)}, \text{ giving a } 5\times \text{ margin.}$$

5.4 Structures Subsystem — Deployment Dynamics

Panel deployment is driven by a pre-loaded torsion spring. Treating each panel as a 50 g uniform rod, $L = 10$ cm:

$$I = \frac{1}{3}mL^2 = \frac{1}{3}(0.05)(0.10)^2 = 1.67 \times 10^{-4} \text{ kg m}^2$$

Spring energy: $E = \frac{1}{2}kx^2 = \frac{1}{2}(5)(0.05)^2 \approx 6.3$ mJ. Peak angular velocity (no damping):

$$\omega_{\max} = \sqrt{\frac{2E}{I}} = \sqrt{\frac{0.0126}{1.67 \times 10^{-4}}} \approx 8.7 \text{ rad/s}$$

Time to 90° ($\pi/2$ rad): $t \approx \pi/(2\omega_{\text{avg}}) \approx 0.18$ s $\ll$ 3 s requirement (R1.1). The 2U structure ($10 \times 10 \times 22.7$ cm) consists of 3D-printed PLA panels reinforced with carbon-fiber rods; center of mass is designed within 2 cm of geometric center to minimize libration on the zipline.

6 Project Plan and Safety

	February				March				April				May	
	1	2	3	4	1	2	3	4	1	2	3	4	1	2
Hands-on demo experiment brainstorm ideas														
Outline of project proposal and presentation														
In-class, hands-on subsystems labs														
Revised proposal and draft slides														
Final proposal and PDR presentation														
Peer reviews and interim course feedback														
In-class, hands-on testing lab														
Journal of 3 integrated tests on the lawn														
Demo Experiment Flight Day Data Work-up														
Individual and Team meetings with CI Instructors														
Class Participation														

The Gantt schedule sequences design, procurement, fabrication, integration, subsystem validation, and end-to-end demo. Critical-path tasks are: component procurement (LoRa radios, HC-SR04P primary + HC-SR04 backup, HB100), 2U structure fabrication in SolidWorks and FDM printing, and radio integration testing.

Safety controls include: servo power removed until actively required for deployment tests; CubeSat carriage tethered except during active runs; all Li-ion battery cycling conducted in a fire-retardant enclosure. An emergency stop procedure (power disconnect switch on CubeSat exterior) is verified functional before each zipline run.

7 Work Breakdown Structure

Table 5: Two-Level Work Breakdown Structure

Subsystem	Task Description	Lead
Electronics	Design power routing (buck converter); wire HC-SR04P, HB100, Hall sensor; continuity and functional testing; telemetry bus validation.	Z. Williams
Software	Pi Pico firmware: 50 Hz acquisition, 5-point baseline subtraction, SD logging, LoRa packetization, contingency mode for link loss.	J. Cavazos
Structure	SolidWorks 2U CAD; torsion spring and hinge analysis; SolidWorks FEA at $2\times$ zipline acceleration; PLA/carbon-fiber fabrication; buried target construction.	J. Goldstein
Communications	Configure dual RFM95W radios; validate link budget at 915 MHz; tune monopole antennas; ground-station Python scripts (packet counter, RSSI log, SD recovery).	A. Beleznay
Testing	Bench, integration, and zipline tests; data reduction; fault-injection verification; risk mitigation review.	All

8 Individual Skill Plans

8.1 Zaire Williams — Electronics and Integration

Zaire will deepen expertise in mixed-signal circuit layout, power rail debugging, and hardware-software interfacing. The power distribution network for this project – buck converters, Schottky OR-ing, per-rail fusing – mirrors spacecraft avionics integration practice directly. A concrete milestone is independently verifying 5 V and 3.3 V rail isolation under load using a bench oscilloscope, and characterizing the Hall-effect sensor output curve before integration. **Peer learning:** Zaire will run a 1-hour bench session with all team members covering power rail diagnostics (reading voltage under load, identifying ground loops, multimeter continuity checks) – ensuring every member can diagnose a wiring fault during integration.

8.2 Jorge Cavazos — Flight Software and Testing

Jorge will strengthen embedded firmware skills through event-driven Python/C code on the Raspberry Pi Pico coordinating multi-sensor acquisition and LoRa packet transmission. Key learning goals: interrupt-driven timing at 50 Hz, sequence-numbered packets for loss detection, and implementing the baseline subtraction algorithm. **Peer learning:** Jorge will pair with Andras during all radio link tests to explain the packet structure so Andras can independently read ground-station logs. Jorge will also brief Zaire on the GPIO timing constraints relevant to pull-up resistor selection on the sensor lines.

8.3 Jonah Goldstein — CAD and Mechanical Systems

Jonah will master SolidWorks for 2U CubeSat structure and deployable mechanism design, including hinge tolerance stack-up and SolidWorks FEA validation of the zipline load case. Designing the torsion spring mount and hinge alignment within the 2U envelope provides direct

practice in constrained mechanism design. **Peer learning:** Jonah will walk the full team through the CAD assembly before any fabrication begins – ensuring all members understand panel geometry, hinge direction, and assembly sequence – and will co-review harness routing with Zaire to prevent cable clearance issues at the hinges.

8.4 Andras Beleznyay — Communications and Mission Operations

Andras will gain hands-on RF testing experience through RFM95W antenna tuning, RSSI characterization at multiple distances, and writing the Python ground-station logging and recovery scripts. He will develop an operational mindset by authoring and executing formal test procedures for Phases I and III. **Peer learning:** Andras will train all team members on the ground-station software interface before the first zipline run – simulating real CubeSat mission operations where multiple ground-station operators must independently monitor and archive telemetry.

9 Resource Support Requests

9.1 Testbed Configuration

One zipline (22 m length, ~1.5 m height) with permission to place a foam-layer target array directly beneath the flight path. Target array: layered copper foil and foam at 5 cm and 10 cm depth, enclosed in a foam-lined sandbox or board frame. Camera mounted below the path for deployment documentation is requested but not required.

9.2 Purchasing and Materials

Approximate total: \$160–220.

- Telemetry: 2× Adafruit RFM95W LoRa transceiver @ \$29.95 (CubeSat + ground station); 3× spring monopole antennas @ \$0.95 each (2 for CubeSat, 1 for ground station).
- Data storage: 1× Adafruit Micro SD SPI breakout @ \$3.50.
- Payload: HC-SR04P ultrasonic sensor (primary, 3.3V–5V compatible); HC-SR04 ultrasonic sensor (backup, requires echo-pin voltage divider); HB100 Doppler radar module (provided by course).
- MCU: Raspberry Pi Pico.
- Power: 1200 mAh 7.4 V Li-ion battery, LM2596 buck converters, AMS1117 LDOs, Schottky diodes.
- Mechanisms: torsion springs (primary), servo motor (backup), Hall-effect sensor.
- Structure: PLA filament, carbon-fiber rods, foam board, copper tape (targets).

9.3 Training and Access

- Soldering — MIT Electronics Research Lab.
- FDM 3D printing — MIT MakerWorkshop.

Fabrication will use the MIT Electronics Research Lab; assembly and integration will use the Unified Lounge; CAD work will use student CAD facility workstations.

10 Conclusion

The ASTEROIDS project bridges 2U CubeSat hardware integration with a high-value planetary-science application: mapping subsurface water ice on Jupiter L5 Trojan asteroids – a resource NASA’s 2023–2032 Decadal Survey identifies as critical for sustainable deep-space exploration [1]. Our zipline demonstration validates the three technical pillars of any future small-body subsurface survey: reliable autonomous instrument deployment, dual-modality subsurface sensing, and robust data return under contingency.

The demonstration is grounded in quantitative subsystem analyses: a 95 dB RF link margin at 915 MHz, a $5\times$ power budget margin (7.5 h runtime), and a spring-driven deployment time of ≈ 0.18 s against a 3 s requirement – all tracing directly to requirements R1–R8 and the Risk Register. Each risk is paired with a concrete fault-injection test that will be executed on the zipline.

Educationally, ASTEROIDS is structured so that every team member achieves both depth in their primary subsystem and breadth through structured peer-learning sessions, cross-subsystem integration reviews, and shared ground-station operations on flight day – directly fulfilling the 16.831 Design Track’s peer learning and hands-on skill development objectives.

References

1. National Academies of Sciences, Engineering, and Medicine, *Origins, Worlds, and Life: A Decadal Strategy for Planetary Science and Astrobiology 2023–2032*, The National Academies Press, Washington, D.C., 2022.
2. D. S. Lauretta et al., “OSIRIS-REx: Sample Return Mission to Asteroid Bennu,” *Space Science Reviews*, 2017.
3. S. Watanabe et al., “Hayabusa2 arrives at the carbonaceous asteroid 162173 Ryugu – a sub-km rubble pile,” *Science*, 2019.
4. N. G. Leveson, *Engineering a Safer World: Systems Thinking Applied to Safety*, MIT Press, Cambridge, MA, 2011.
5. W. L. Stutzman and G. A. Thiele, *Antenna Theory and Design*, 3rd ed., John Wiley & Sons, Hoboken, NJ, 2012.
6. NASA Jet Propulsion Laboratory, *810-005: DSN Telecommunications Link Design Handbook*, JPL, Pasadena, CA, 2023.
7. P. Michel et al., “The ESA Hera Mission: Detailed Characterization of the DART Impact Outcome and of the Binary Asteroid (65803) Didymos,” *The Planetary Science Journal*, 2022.
8. Semtech Corporation, *SX1276/77/78/79 Datasheet: 137 MHz to 1020 MHz Long Range Low Power Transceiver*, Semtech.
9. MIT AeroAstro 16.831 Design Track, Assignment DT-2 and PDR materials, Spring 2026.